

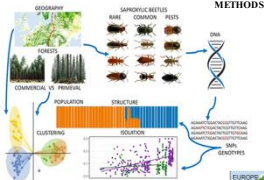
Drivers of genetic diversity in keystone forest species: contrasting responses of rare and common saproxylic beetles to habitat location and naturalness

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INTRODUCTION

Saproxylic beetles are crucial for forest ecosystem functioning, yet knowledge of their population genetics remains limited. In particular, the relationships between geographic isolation, forest quality, and management practices and the genetic structure and diversity of species with different ecological statuses (e.g., primeval forest relicts or timber pests) are still poorly understood. Such information is essential for understanding the evolutionary history of these species and for effective conservation planning and forest management. Here, we address these questions using SNP data from several species and environmental information from selected forests in Central Europe, an important transitional zone in both biogeography and forest management practices.



METHODS

1. Approach: Multi-species Double Digest RAD-seq (ddRAD).
2. Scale: 600 km (W-E) and 500 km (S-N) across Central Europe.
3. Sample: 12 beetle species spanning varying ecological tiers.
4. Comparison: Commercial stands vs. Primeval protected forests.

SAMPLING



KEY RESULTS

- Forest management and geography both shape saproxylic beetle population genetics.
- Most species showed isolation by distance patterns.
- Forest management had a stronger effect on genetic differentiation (F_{ST}) than geographic distance alone.
- Rare and old-growth specialist taxa retained higher genetic distinctiveness in primeval forests.
- Commercial forest stands supported lower genetic uniqueness of specialist species.
- Generalist and pest species were comparatively resilient to intensive logging.
- Preserving primeval forests is critical for conserving the genetic heritage of specialized saproxylic beetle communities.

CONSERVATION AND MANAGEMENT IMPLICATION

- **Define Evolutionary Units:** Strong geographic structuring suggests that regional beetle populations should be managed as distinct conservation units to preserve unique genetic lineages.
- **Protect Genetic Reservoirs:** Old-growth and primeval forests are essential genetic sources; protecting these "islands" is critical for the survival of habitat-specialist species.
- **Enhance Connectivity:** Management must prioritize landscape-scale connectivity to facilitate gene flow between commercial stands and protected old-growth remnants.
- **Precision Monitoring:** Genetic data should be integrated into monitoring programs to identify and protect populations at risk of isolation or low genetic diversity.

CONCLUSION

To preserve beetle biodiversity, conservation efforts must prioritize regional connectivity and the protection of primeval refugia, as these are essential for maintaining the evolutionary potential of forest specialists.

